Representing Functions and Fields

Finitely many numbers for a function, and what each choice loses

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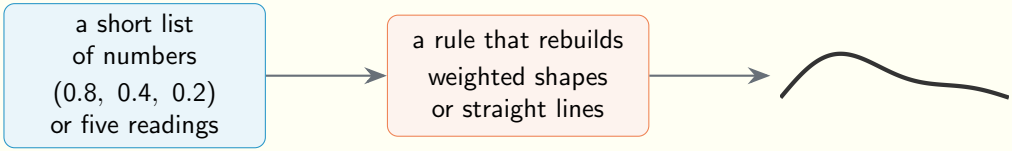
Representing a function as a discrete list and a rule

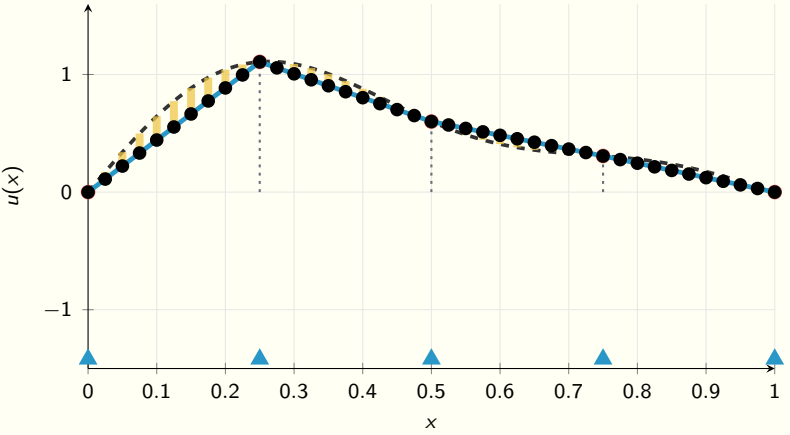


How well can finitely many numbers stand in for a function, and what decides the accuracy?

Outline

1. Readings at fixed stations
2. Coefficients on shapes
3. Norms and the meaning of error
4. Inner products, projection, and orthogonality
5. Changing the basis
6. Convergence, density, and completeness
7. Kernels and point evaluation
8. Noisy observations, ridge regression, and Gaussian conditioning
9. Approximating a family of functions
10. Adaptive representations
11. Choosing a representation for the cooling fin
12. From hats to a network

Sampling the displacement at fixed stations



five readings

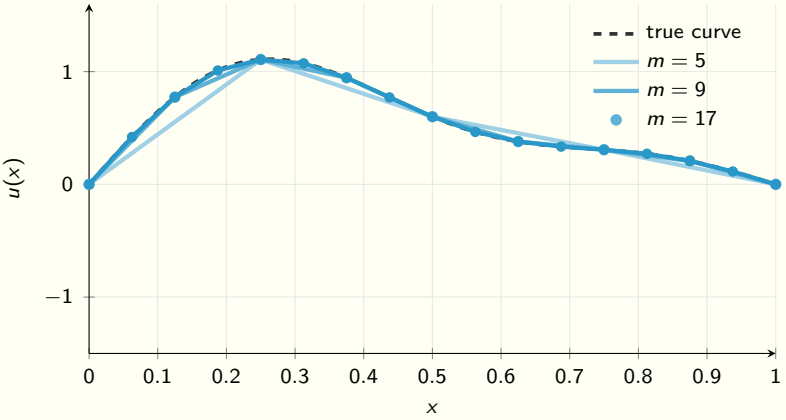
0, 1.11, 0.60, 0.31, 0

worst miss, five sensors

0.22, **or** 20.1 **percent**

Try it live

Convergence of piecewise-linear interpolation with the spacing



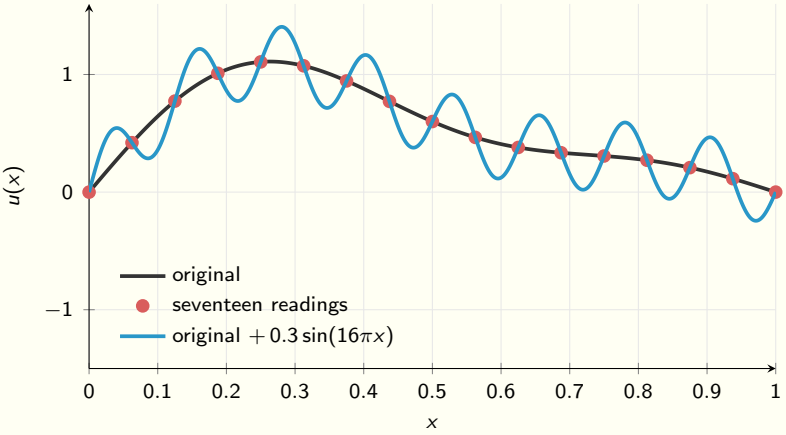
$m = 5,\ h = 1/4$
20.1 percent

$m = 9,\ h = 1/8$
6.3 percent, a factor of 3.2

$m = 17,\ h = 1/16$
1.6 percent, a factor of 4.0

slope on log axes
-2.07

Aliasing of a wave that vanishes at every station



change in the seventeen readings

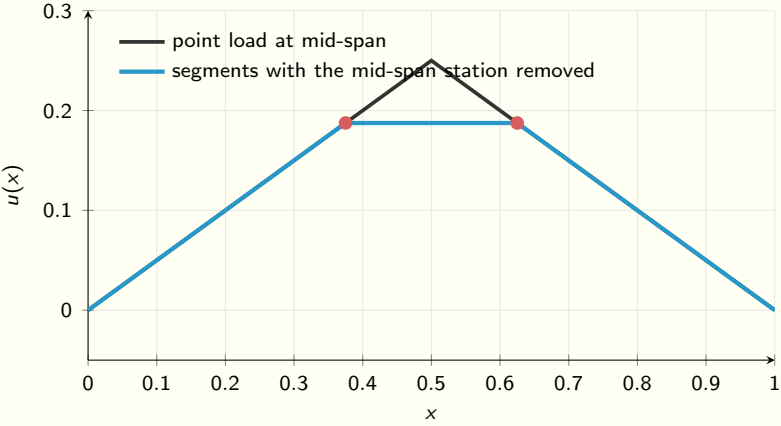
0

change at each crest

0.3, **about a quarter of the largest displacement**

Try it live

Grid dependence of sampled representations



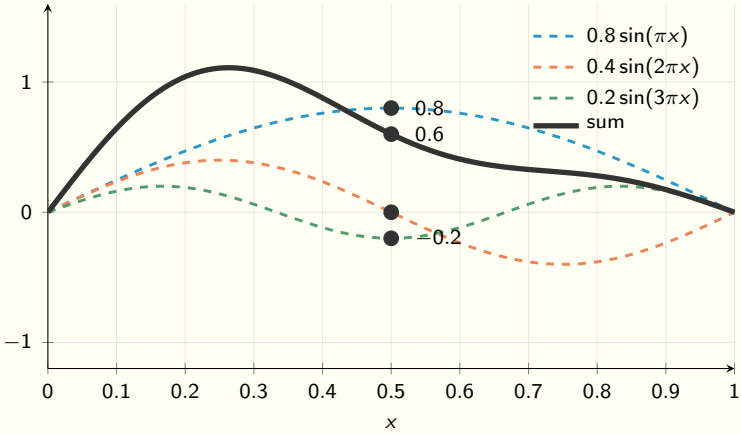
corner, mid-span sensor removed

worst miss 0 **to** 12.5 **percent**

smooth displacement, same sensor removed

1.6 **to** 1.7 **percent**

Representing a function with coefficients on sine modes



the whole curve

three numbers,

$(0.8, 0.4, 0.2)$

check at mid-span

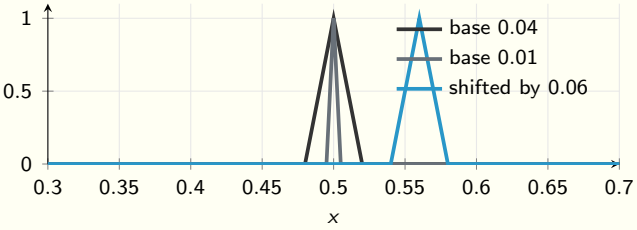
$u(\frac{1}{2}) = 0.8 - 0.2 = 0.6$

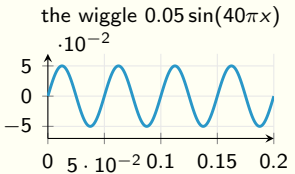
Try it live

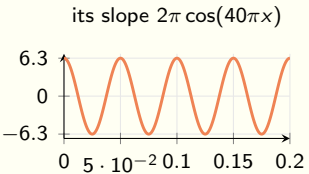
The coe cient list as coordinates in a vector space

$$u + v = \sum_{i=1}^{3} (a_i + b_i)\ _i ; \qquad u = \sum_{i=1}^{3} (a_i)\ _i$$

Disagreement between the sup, L^2, and H^1 norms







spike, base 0.04
sup 1, L^2 0.115, **energy** 50

spike, base 0.01
sup 1, L^2 0.058, **energy** 200

L^2 distance, spike to shifted spike
$\sqrt{2} \times 0.115 = 0.163$

wiggle $0.05\sin(40\pi x)$
sup 0.05, L^2 0.035, H^1 4.44

Try it live

Definition of a norm and a seminorm

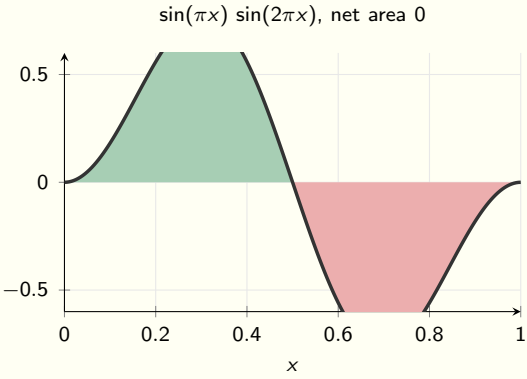
Norm

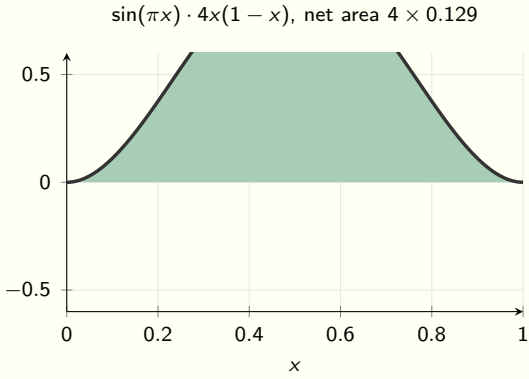
$$\|u\| = 0 \text{ only for } u = 0, \qquad \|\alpha u\| = |\alpha|\,\|u\|, \qquad \|u + v\| \le \|u\| + \|v\|.$$

Seminorm

$$|c|_{H^1} = \left(\int_0^1 (c')^2\,dx\right)^{1/2} = 0 \quad \text{for every constant } c, \qquad \text{and on the pinned string the only const}$$

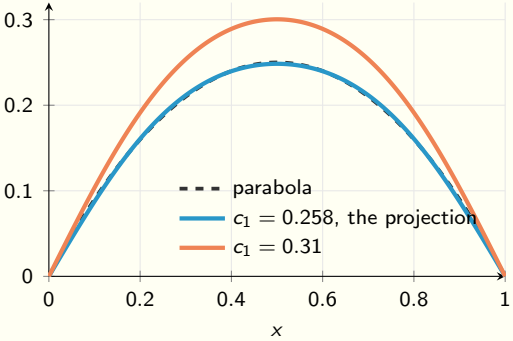
The inner product as signed overlap

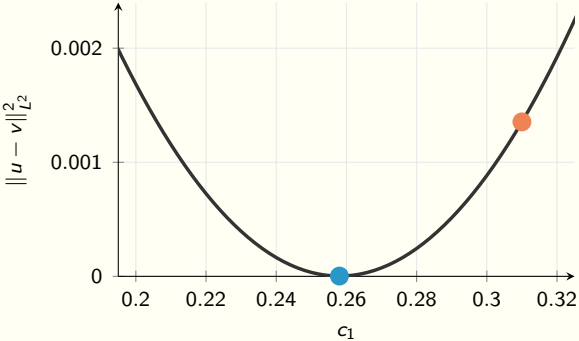




$$\langle f, g \rangle = \int_0^1 f(x)\, g(x)\, dx$$

Minimizing squared error via orthogonal projection



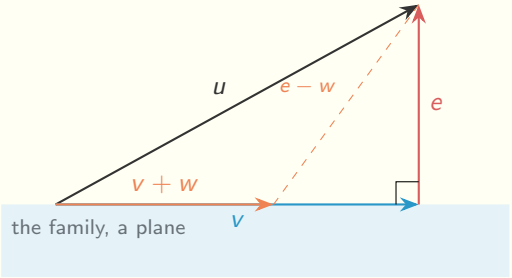


$c_1 = 8/\pi^3 = 0.258$
error 0.87 **percent**

$c_1 = 0.31$, twenty percent high
error norm 23 **times larger**

Try it live

Uniqueness of the projection and the Pythagorean identity



$$\|e - w\|^2 = \|e\|^2 + \|w\|^2$$

$$\|u\|^2 = \|v\|^2 + \|e\|^2, \qquad 0.033333 = 0.033331 + 2$$

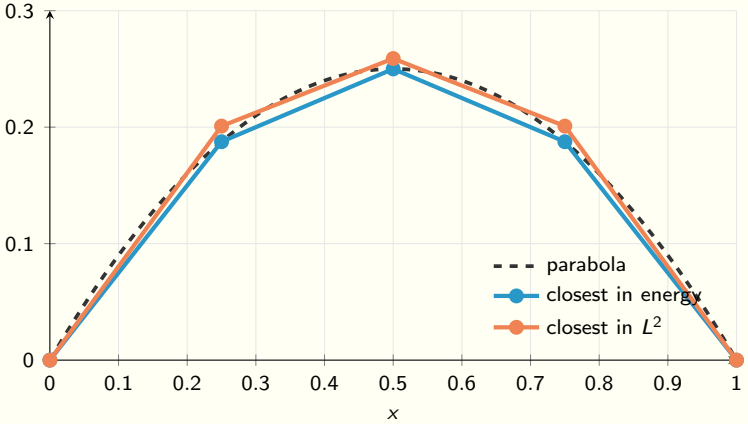
The projection theorem

Projection theorem

In a space with an inner product, a finite-dimensional family has one closest member to each target, and that member is the one whose error is orthogonal to the whole family.

$\langle u - v, \phi_j \rangle = 0$ for every shape ϕ_j, $\quad c_k = \dfrac{\langle u, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle}$ when the shapes are orthogonal.

Projection in the L^2 and energy inner products



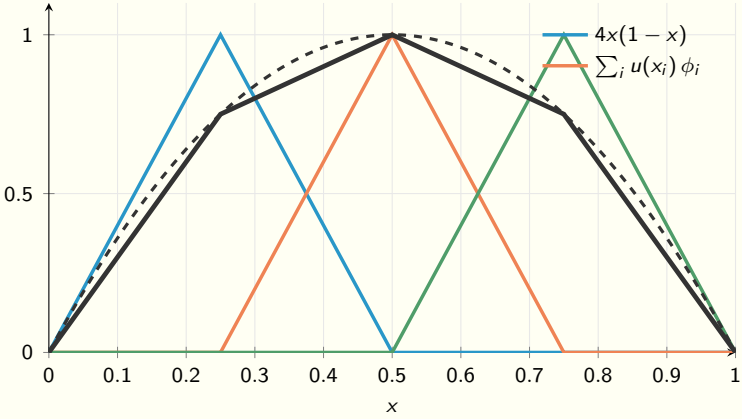
closest in energy, heights
0.1875, 0.25, 0.1875

closest in L^2, heights
0.2009, 0.2589, 0.2009

L^2 error, L^2 against energy
3.3 **against** 6.3
percent

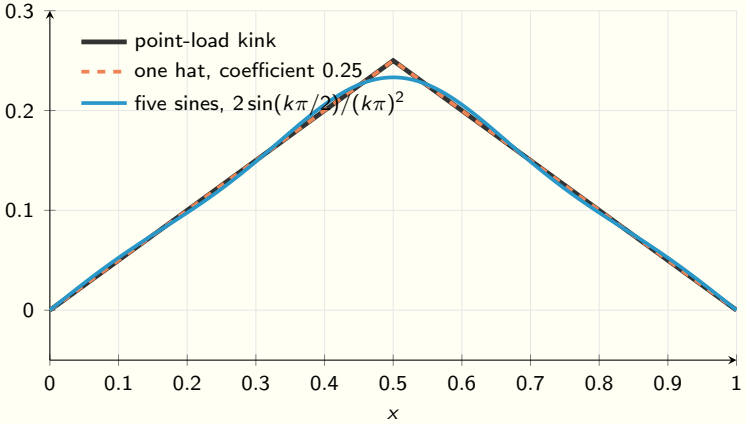
H^1 error, L^2 against energy
26.0 **against** 25.0
percent

Piecewise-linear interpolation as a hat-function expansion



$$u_h(x) = \sum_i u(x_i)\,\phi_i(x)$$

Representing a kink in the hat basis and the sine basis



one hat on the corner
one number, 0.25**, no error**

three hats, five stations
0.125**,** 0.25**,** 0.125**, no error**

sines, three and thirty-three modes
4.8 **and** 0.20 **percent**

Try it live

Decay rates of sine coe cients

load f = 2

$c_k = 8=(k\ ),$ slope 1:0

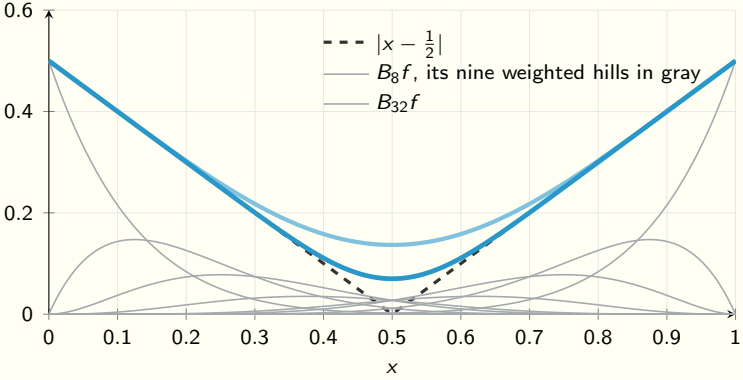
kink

$c_k = 2=(k\ )^2,$ slope 2:0

parabola

$c_k = 8=(k\ )^3,$ slope 3:0

Convergence rate of Bernstein polynomials



$$B_n f(x) = \sum_{k=0}^{n} f\left(\tfrac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

gap at the corner, $n = 2$
0.250

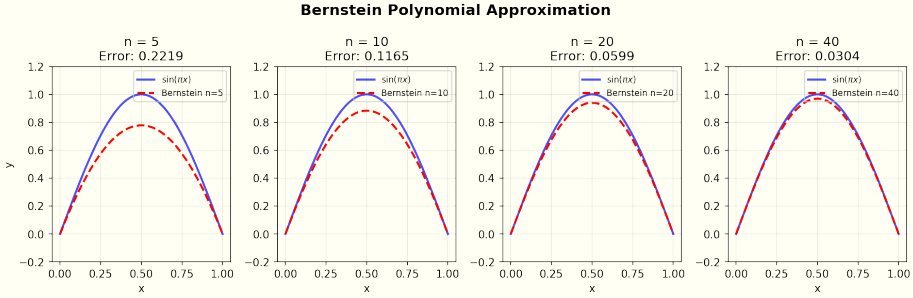
gap at the corner, $n = 8$
0.137, **predicted**
0.141

gap at the corner, $n = 32$
0.070, **predicted**
0.071

prediction
$0.4/\sqrt{n}$

Try it live

Bernstein basis polynomials

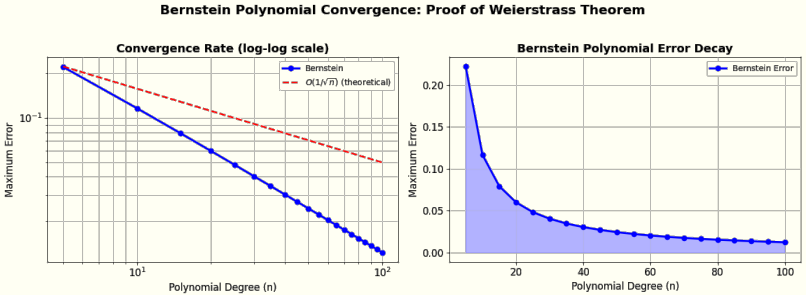


$$B_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}, \qquad B_n f(x) = \sum_{k=0}^{n} f\left(\tfrac{k}{n}\right) B_{k,n}(x)$$

largest gap,
$n = 5, 10, 20, 40$
0.222, 0.117, 0.060, 0.030

the basis
nr3 d4G0 g 0 ne he basB.

Convergence of Bernstein polynomials for a smooth function

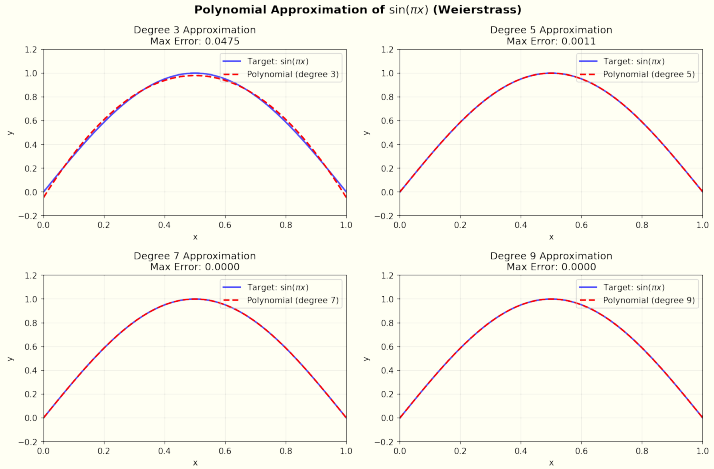


target
$\sin(\pi x)$

largest gap, $n = 5$ and $n = 100$
0.22 **and** 0.013

Voronovskaya at mid-span, $\pi^2/(8n)$
0.25 **and** 0.012

Least-squares polynomial approximation of $\sin(\pi x)$



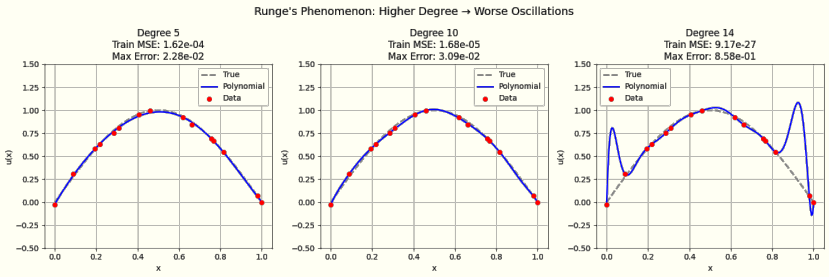
degree 3, largest gap

0.0475

degree 5, largest gap

0.0011

Runge's phenomenon



degree 5, least squares
largest gap 0.023

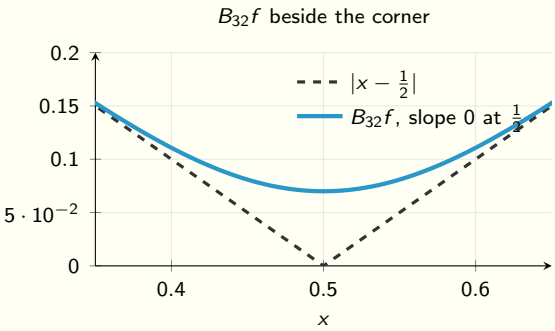
degree 10, least squares
largest gap 0.031

degree 14, through all fifteen readings
training mean squared error 9×10^{-27}**, largest gap** 0.86

The Weierstrass approximation theorem and its limits

Weierstrass (1885)

For every continuous f on $[0,1]$ and every tolerance $\epsilon > 0$ there is a polynomial p with $|f(x) - p(x)| < \epsilon$ at every point of the interval.

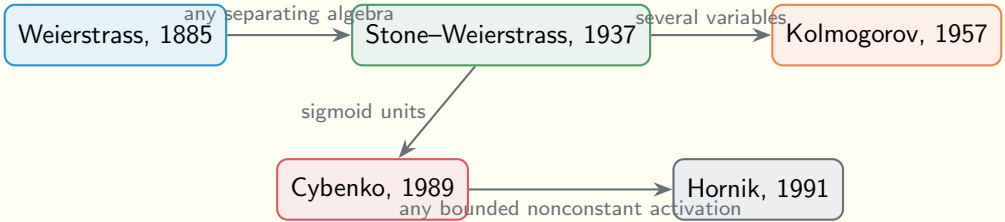


Bernstein's error
falls like $1/\sqrt{n}$

best degree-n polynomial
falls like $1/n$

the theorem names
no polynomial, no degree, no slope

History of universal approximation theorems

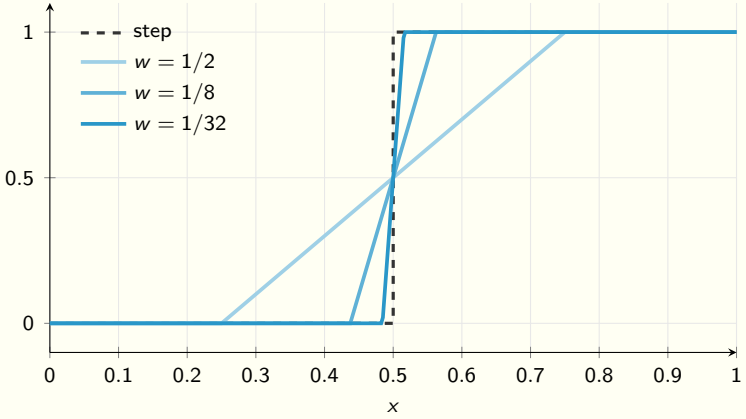


what every theorem on this line gives

existence for any tolerance

what none of them3 T 0-6 69217 9 55456378 q1m3 T

Cauchy sequences, completeness, and Hilbert spaces



L^2 distance to the step, $\sqrt{w/12}$

0.204 **at** $w = 1/2$

$w = 1/8$

0.102

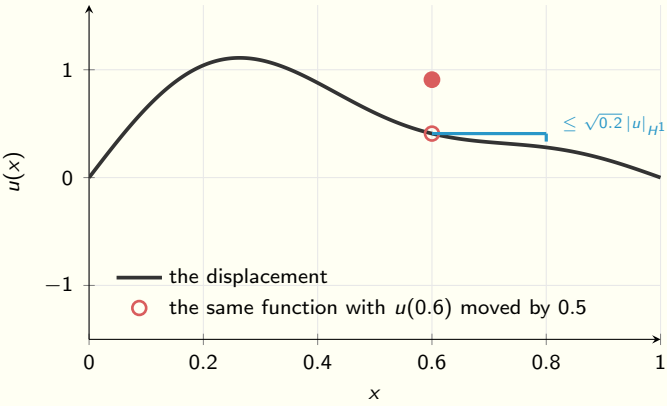
$w = 1/32$

0.051

$\|f_8 - f_{32}\|_{L^2}$

0.

Point evaluation in L^2 and H^1

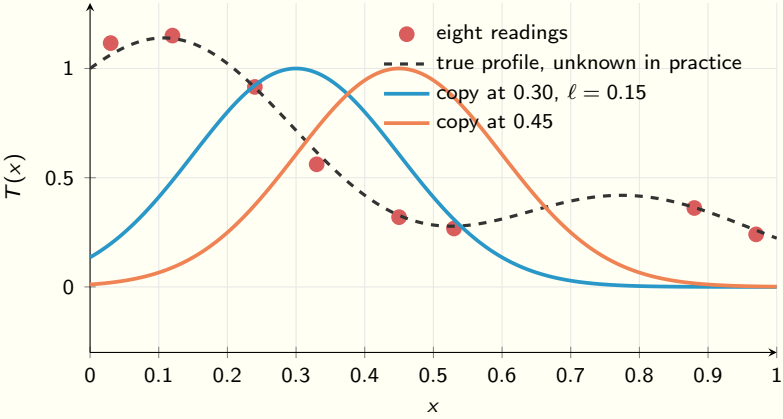


L^2 distance between the two

0

$$|u(y)-u(x)| = \left| \int_x^y u'(s)\, ds \right| \leq \sqrt{}$$

Kernels as similarity rules for scattered readings



$$k(x, x') = e^{-(x-x')^2/2\ell^2}$$

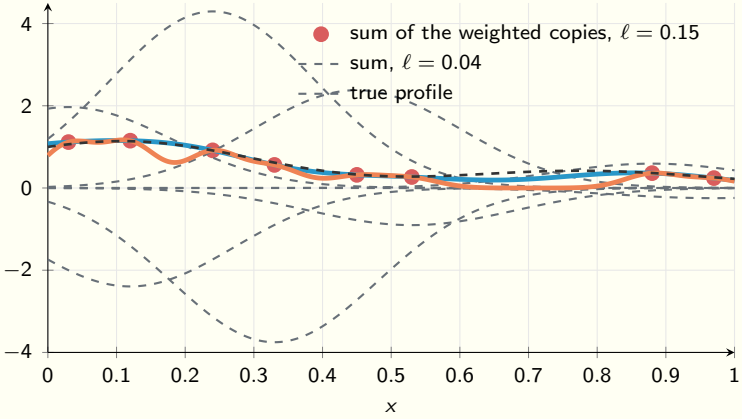
$k(0.30, 0.45)$, one ℓ apart

0.61

k at 3ℓ apart

0.011

Reconstructing a profile with weighted kernel copies



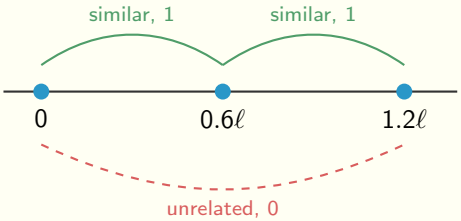
$$\mathbf{K}\boldsymbol{\alpha} = \mathbf{y}, \qquad K_{ij} = k(x_i, x_j)$$

weights at $\ell = 0.15$
alternate in sign, up to 4.30 **and** -3.75

two equal readings one ℓ apart
0.62 **each**

sum at $x = 0.7$, $\ell = 0.04$
0.000 **against the true** 0.393

Positive definiteness of a kernel

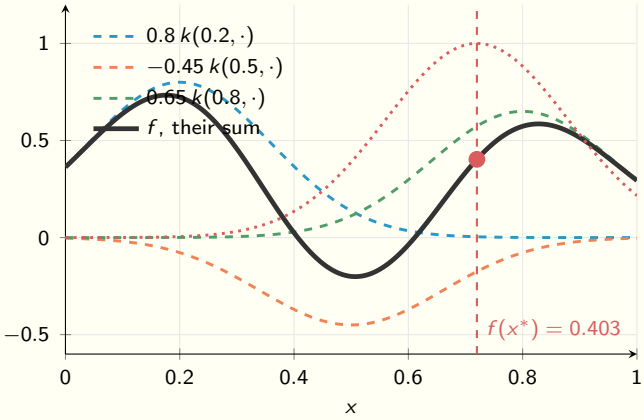


$$\mathbf{w} = (1, -\sqrt{2}, 1), \qquad \mathbf{w}^\top \mathbf{K} \mathbf{w} = -1.66$$

$$k(x, x') = \phi(x)^\top \phi(x') \;\Rightarrow\; \mathbf{w}^\top \mathbf{K} \mathbf{w} = \Big\| \sum_i w_i$$

$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \lambda = 1 - \sqrt{2},\ 1,\ 1 + \sqrt{2}$$

The reproducing kernel Hilbert space



$$\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$$

$\|f\|_k^2 = \boldsymbol{\alpha}^\top \mathbf{K} \boldsymbol{\alpha}$, $\ell = 0.16$

1.041

$$f(x) = \langle f, k(x, \cdot) \rangle_k$$

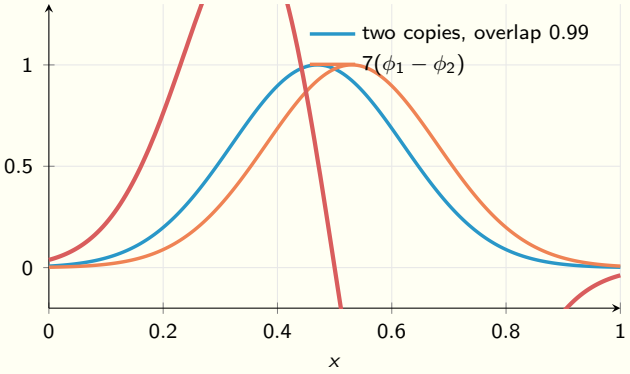
$f(0.72)$, read as an inner product

0.403, **bound** 1.02

slope bound

$|f'(x)| \leq \|f\|_k / \ell$

Noise amplification in kernel interpolation



eigenvalues of $\mathbf{K}$, overlap 0.99

1.99 **along** $(1, 1)$**,** 0.01 **along** $(1, -1)$

readings $(1, 1)$

weights near $\frac{1}{2}$ **each**

noise $(\varepsilon, -\varepsilon)$

weights $\pm\varepsilon/0.01$**, a hundred times the noise**

$\|\phi_1 - \phi_2\|$

0.14**, so the curve moves fourteen times the noise**

Kernel ridge regression and the choice of λ

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^{n} \big(f(x_i) - y_i\big)^2 + \lambda \|f\|_k^2$$

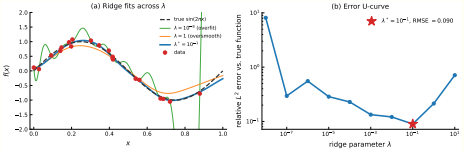
$$(\mathbf{K} + \lambda \mathbf{I})\boldsymbol{\alpha} = \mathbf{y}$$

$\lambda = 0.1$, small divisor

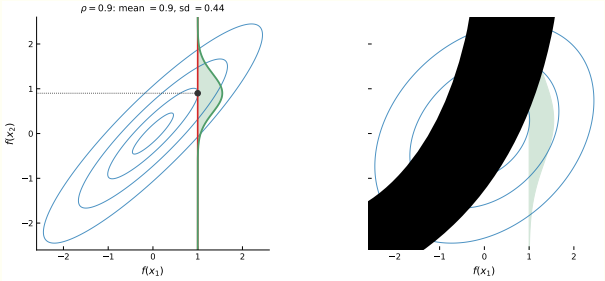
$0.01 \rightarrow 0.11$

amplification

a hundred $\rightarrow$ nine



Conditioning a Gaussian on an observation



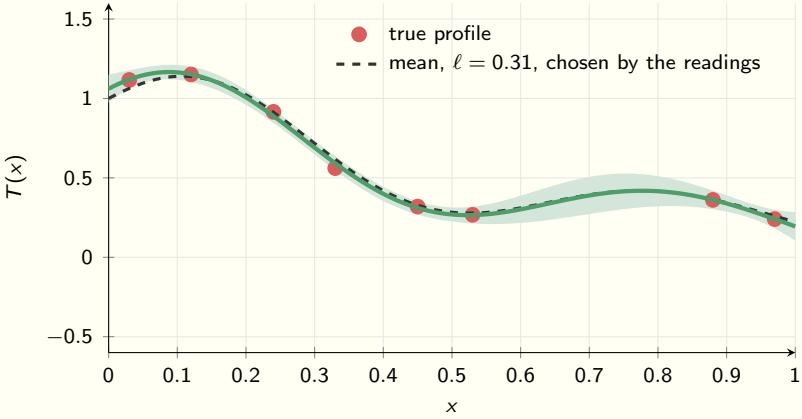
$$f(x_1) = Z_1, \qquad f(x_2) = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2$$

$$f(x_2) \mid f(x_1) = y \;\sim\; \mathcal{N}\left(\rho y,\; 1-\rho^2\right)$$

$\rho = 0.9$
mean $0.9y$**, standard deviation** 0.44

$\rho = 0.3$
mean $0.3y$**, standard deviation** 0.95

Gaussian-process reconstruction of the fin profile



mean

$\mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$

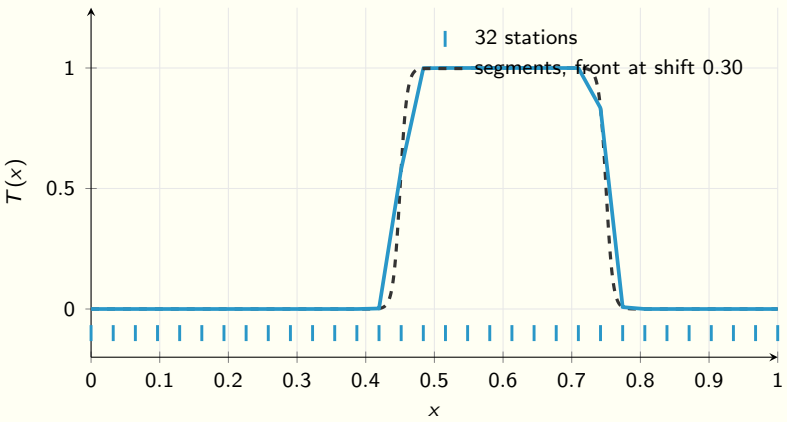
variance

$k(x, x) - \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(x)$

$\ell = 0.31$

error 3.2 **percent,**
band in the gap 0.12

Approximating a family of functions with a fixed basis



32 stations, shift 0
8.7 **percent**

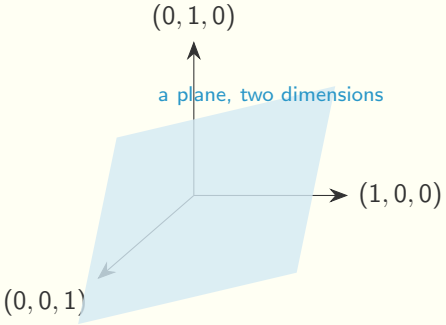
32 stations, shift 0.15
6.3 **percent**

32 stations, shift 0.30
9.6 **percent**

station spacing
three times the edge width

numbers that rebuild the front at every time
two edges, a width, an amplitude, a shift

The Kolmogorov n-width



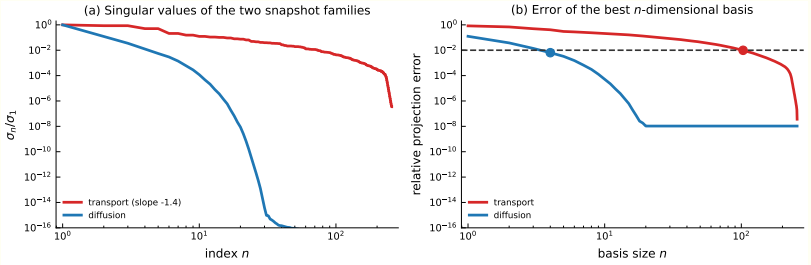
Kolmogorov n-width

$$d_n(\mathcal{M}) = \min_{\dim V = n} \max_{f \in \mathcal{M}} \min_{v \in V} \|f - v\|$$

three unit spikes, squared distances to any plane **sum to** $3 - 2 = 1$

worst spike **at least** $\sqrt{1/3} = 0.577$

Widths of diffusion and transport families



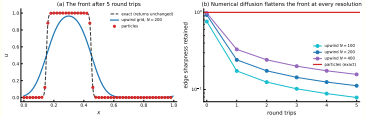
diffusing pulse, one percent
four modes

translating pulse, one percent
103 **modes**

coefficient k under diffusion
shrinks by $e^{-(2\pi k)^2 t}$

coefficient k under translation
changes phase only

Adaptive representations with particles, splats, and selection

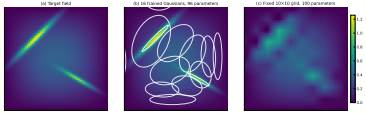


markers after five round trips

front unchanged

upwind grid, edge steepness kept

eight to sixteen percent

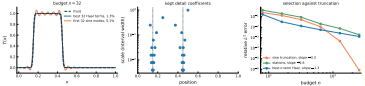


sixteen trained Gaussians, 96 parameters

error 0.0125

fixed grid, 100 coefficients

error 0.30



thirty-two largest Haar coefficients

1.3 **percent**

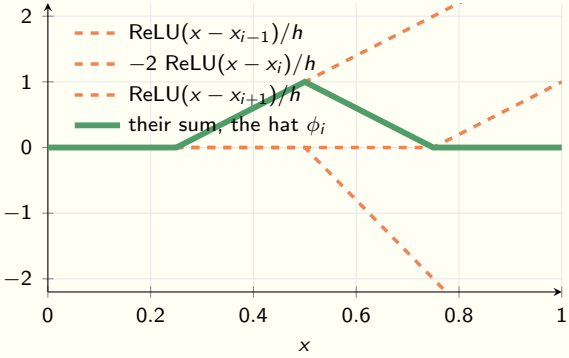
first thirty-two sines

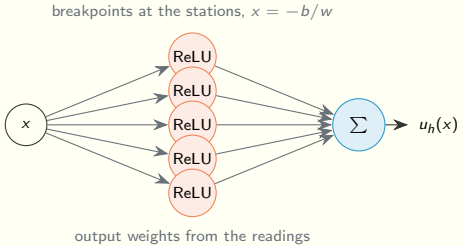
5.1 **percent**

Choosing a representation for the cooling n

description	error vs truth	held-out
straight segments	6:5 percent	0:095
cubic, least squares	23	0:310
degree-7 interpolation	31	3:57
kernel interpolant, ` = 0:15	11	0:112
Gaussian-process mean, ` = 0:31	3:2	0:034
noise level		0:03

Piecewise-linear interpolation as a one-hidden-layer ReLU network



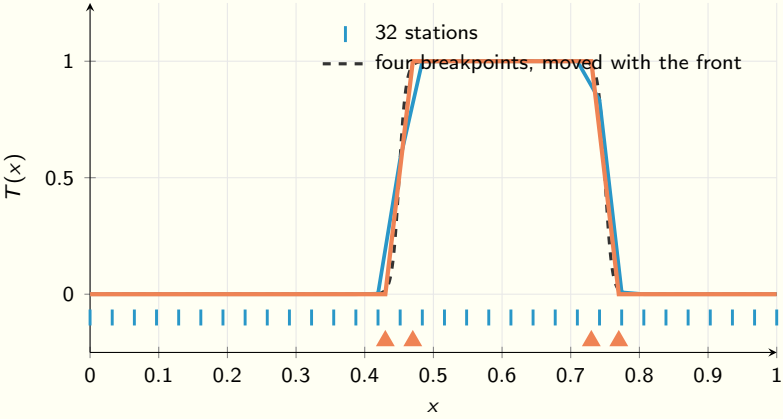


bound $Mh^2/8$ at five stations

0.0156, **met with equality by the parabola**

Try it live

Movable ReLU breakpoints on a traveling front



breakpoint of ReLU$(wx + b)$

$x = -b/w$

32 stations, front shifted by 0.3

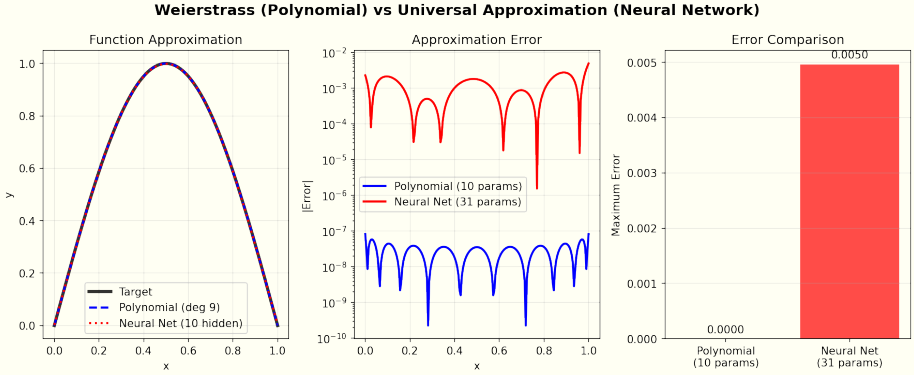
9.6 **percent**

four breakpoints, moved with the front

4.9 **percent**

Try it live

Polynomial and neural network approximation of $\sin(\pi x)$



polynomial, ten numbers

largest gap below 10^{-7}

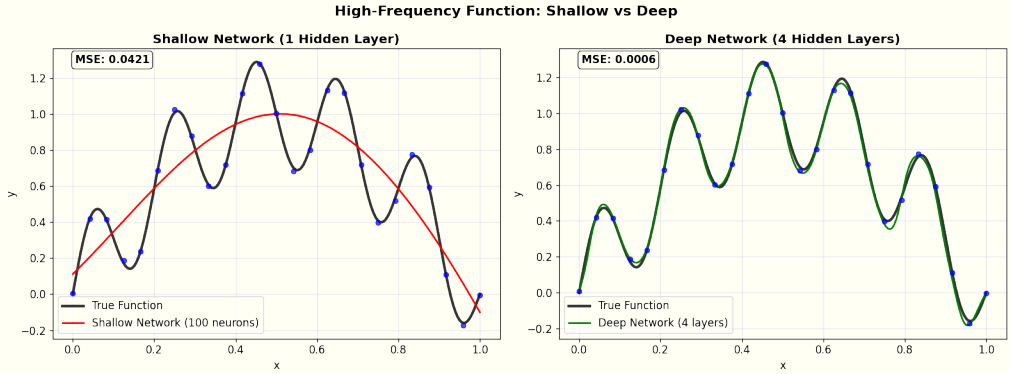
network, thirty-one numbers

largest gap 0

Polynomials and neural networks compared

	polynomials	one-hidden-layer networks
shapes	global, every coefficient changes the curve everywhere	local ramps, each unit changes the curve beyond its breakpoint
placement	fixed by the degree	breakpoints are trained parameters
failure on sparse data	Runge oscillations near the ends	a descent that stalls
growth with dimension	coefficients grow like $\binom{n+d}{d}$ in d dimensions	parameters set by width and depth
structure	one sum	layers compose

Shallow and deep networks on a high-frequency function



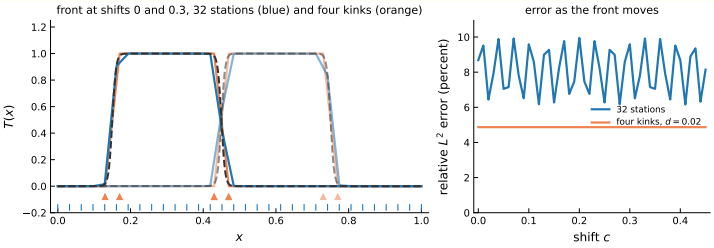
one hidden layer, a hundred units

mean squared error 0.0421

four hidden layers

mean squared error 0.0006

From fixed shapes to trainable breakpoints



fixed breakpoints

a linear solve, with the projection theorem's guarantee

moving breakpoints

gradient descent, with no guarantee

Summary

1. Readings at fixed stations
2. Coefficients on shapes
3. Norms and the meaning of error
4. Inner products, projection, and orthogonality
5. Changing the basis
6. Convergence, density, and completeness
7. Kernels and point evaluation
8. Noisy observations, ridge regression, and Gaussian conditioning
9. Approximating a family of functions
10. Adaptive representations
11. Choosing a representation for the cooling fin
12. From hats to a network